1 Background & Motivation

Real-time human pose estimation has made it possible to monitor resistance-training exercises using a single, already-installed camera — with no wearable sensors, no depth camera, and no multi-view rig. Systems built on a MediaPipe-based pose pipeline combined with a Temporal Convolutional Network (TCN) for exercise classification and RepNet for class-agnostic repetition counting have shown that this setup can reliably identify which exercise is being performed and count repetitions in real time, on ordinary hardware, using footage from a real gym rather than a controlled studio. However, such systems are **form-agnostic**: every detected repetition is counted identically, regardless of whether it was executed with correct technique.

1.1 Resistance Training Injury Rates & Mechanisms

- **Epidemiology of Injury:** Epidemiological studies indicate that resistance training and bodybuilding yield an injury incidence rate ranging between **2.4 to 5.3 injuries per 1,000 hours of training**.
  - **Individual Lifter Context:** Training 4 hours/week equals ∼200 hours/year. This rate implies that a regular lifter will experience **at least one reportable injury every 2 to 2.5 years** due to improper movement mechanics. Heavy lifting
  - **Facility Context:** For a medium gym (50 active lifters), this translates to 25 to 50+ injuries per year across the facility, primarily driven by poor kinematics during heavy compound lifts like Deadlifts and Squats.
- **Primary Causes:** Improper execution kinematics, lack of exercise knowledge, and lifting uncontrolled loads without adhering to correct biomechanical joint angles account for over **50% of acute and overuse gym injuries**.
- **High-Risk Anatomical Sites:** Biomechanical evaluations show that the **lower back (lumbar spine)**, **shoulders**, and **knees** account for more than **65% of all recurring weightlifting injuries**. These stem directly from classic form errors such as:
  - *Spinal Rounding* during deadlifts and squats.
  - *Knee Valgus* (inward collapsing of knees) during squat phase transitions.
  - *Excessive Elbow Flaring* during bench presses and pushups.

1.2 Member Retention and Supervision Deficit

- **Gym Dropout Rates:** Surveys show that approximately **60% of new gym members drop out within their first six months of membership**.
- **Lack of Supervision:** The primary drivers for dropout are fear of injury, lack of confidence in equipment usage, and the high cost of one-on-one personal trainers.
- **Aversion to Wearables:** Approximately **60% of gym-goers express explicit reluctance toward wearing physical sensors or specialized tracking hardware**, reinforcing the necessity for unobtrusive, camera-based visual feedback solutions.

2 The Gap & Related Work

Systems capable of assessing execution *quality* and explaining specific errors do exist — but each relaxes one or more of the constraints that make single-camera gym monitoring cheap and deployable in the first place:

- **FLEX** (Yin et al., 2025) links biomechanical errors such as knee valgus or spinal rounding to corrective feedback through a Fitness Knowledge Graph, but requires five synchronized

camera views plus surface EMG electrodes.

- **Fitness-AQA** (Parmar et al., 2022) provides expert-graded error labels for real-world weightlifting footage, but only as a binary correct/incorrect grade per clip, with no per-error explanation, and no real-time component.
- **CoT-AFA / Explainable Fitness Assessor** (Qi et al., 2025) generates step-by-step, chain-of-thought explanations of form errors, but has not been evaluated for real-time, single-camera deployment.
- **M3GYM** (Xu et al., 2025) captures genuinely unconstrained, real-gym footage with expert action assessments, but requires eight synchronized camera viewpoints.
- **Physical Rehabilitation Systems:** Single-camera quality *scoring* is well established in clinical domains — **KIMORE** (Capecci et al., 2019) and **UI-PRMD** (Vakanski et al., 2018) both assign a numeric quality score from a single-view or Kinect recording — but only for gentle, clinician-prescribed movements, yielding one coarse/continuous score rather than a structured, named, per-error explanation.
- **Early Prototypes:** Early single-camera, wearable-free form-correction prototypes for gym exercises exist, but their evaluations are limited to controlled home- or lab-like settings and explicitly flag real, crowded-gym conditions as untested.

System	**Single ordinary camera**	**No wearables / EMG**	**Real-time**	**Per-error, explainable feedback**	**Validated in real, unconstrained gym**
Base thesis (Cridlig)	✓	✓	✓	×	✓
Fitness-AQA (Parmar et al., 2022)	∼	✓	×	× (binary grade only)	∼
FLEX (Yin et al., 2025)	× (5 views)	× (sEMG)	×	✓	×
CoT-AFA (Qi et al., 2025)	—	—	×	✓	—
M3GYM (Xu et al., 2025)	× (8 views)	✓	×	∼ (completeness only)	✓
KIMORE / UI-PRMD	✓	∼	×	× (single score, no named error)	× (clinic; different exercise domain)
Proposed system	✓	✓	✓	✓	✓ *(target)*

Table 1: *

✓ = meets the criterion, × = does not, ∼ = partial, — = not reported in the sources reviewed.

No existing system combines all five columns. A further consequence of today's form-agnostic counting is that a reported "10 repetitions" carries no information about how many of those repetitions were actually performed within acceptable form limits.

3 Problem Statement

Traditional vision-based exercise monitoring systems evaluate repetition quantity rather than execution safety, relying on class-agnostic algorithms that treat incorrectly performed repetitions identically to valid ones. While biomechanical form analysis can reduce exercise-induced injuries and ensure training effectiveness, existing automated assessment methods either rely on complex multi-camera setups, specialized wearables, or depth sensors that are impractical for standard gym environments.

Furthermore, current single-camera prototypes primarily provide binary classification (correct vs. incorrect) or continuous numerical scores, lacking real-time multi-level diagnostic capability (error identification, severity rating, and targeted corrective feedback). Consequently, there is currently no real-time system that, using only a single ordinary camera and no wearable or depth sensors, can recognize a resistance-training exercise, count repetitions, detect specific form errors on a per-repetition basis, generate actionable feedback for each detected error, and report a form-validity-aware repetition count (correctly vs. incorrectly executed repetitions) — validated under realistic, unconstrained gym conditions rather than a controlled studio or laboratory setting.

4 Research Questions

- **RQ1 — Recognition & Counting (Foundation):** Building on existing single-camera pipelines, can exercise type and repetition boundaries be identified reliably in real time for the target exercise set, under realistic gym conditions rather than curated footage?
- **RQ2 — Form/Error Detection (Core Contribution):** Can specific, biomechanically defined form errors (e.g., knee valgus, insufficient depth, spinal rounding) be detected on a per-repetition basis from single-camera pose estimates alone — without sEMG or multi-view triangulation — and at what accuracy relative to multimodal systems such as FLEX?
- **RQ3 — Explainable Feedback (Core Contribution):** Can detected errors be translated into specific, actionable, natural-language feedback in real time, rather than a single aggregate correct/incorrect label?
- **RQ4 — Validity-Aware Counting (Core Contribution):** Does reporting repetitions as valid vs. invalid provide a more actionable training signal than a single raw count, and how should per-repetition "validity" be operationally defined?

5 Academic References

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